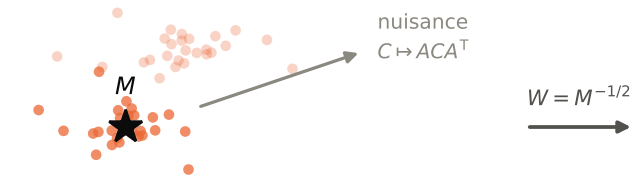
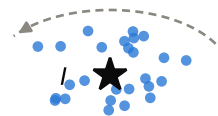
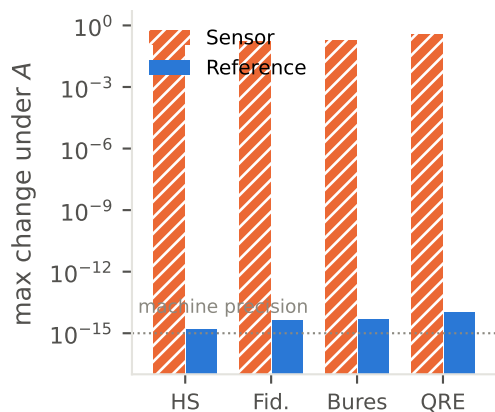
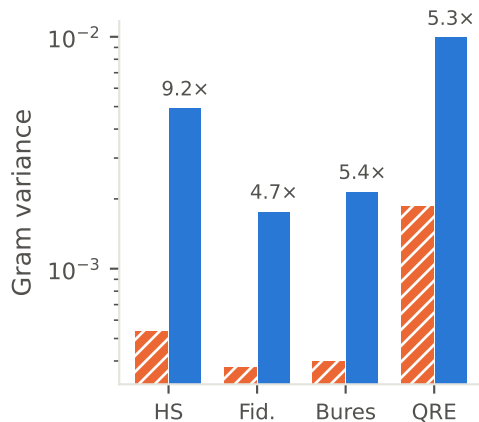


**a****Sensor frame**cluster moves under  $A$ 

$A$  becomes a rotation  $U$ ;  
unitary invariants are unchanged

**Reference frame**

spread, and invariant

**b****c****d**